

CSCE 411

Dynamic Programming Practice

Describe and analyze efficient bottom-up dynamic programming algorithms for the following problems:

1. Given an array $A[1, \dots, n]$ of integers, compute the length of the longest decreasing subsequence of A .
2. Given an array $A[1, \dots, n]$ of integers, compute the length of the longest alternating subsequence of A . We say that a sequence $x_1, \dots, x_k$ is *alternating* if $x_i < x_{i+1}$ for every odd index $1 \leq i \leq k-1$ and $x_i > x_{i+1}$ for every even index $1 \leq i \leq k-1$.
3. Given an array $A[1, \dots, n]$ of integers, compute the length of the longest palindrome subsequence of A . We say that a sequence $x_1, \dots, x_k$ is a *palindrome* if $x_i = x_{k-i+1}$ for every index $1 \leq i \leq k$.
4. You are at a fancy sushi restaurant which uses a massive conveyor belt to display the sushi. You have come prepared to maximize your enjoyment of this dining experience, so you assembled a list of the n different pieces of sushi $s_1, \dots, s_n$ that will be served on the conveyor belt, in the order that they will arrive in front of you. For each piece of sushi, you have recorded:
 - The enjoyment e_i that you will receive from eating s_i (You have found a way to quantify this, don't ask me how)
 - The fullness f_i that you will receive from eating s_i , measured in how many piece of sushi you will need to let pass by before you can eat another (assume that this includes the time it takes you to eat s_i).

Each piece of sushi will only pass by you once, and you must immediately choose whether to eat it or let it pass.

Give an algorithm that takes this information about the sushi restaurant and determines the maximum total enjoyment that you can receive from this dining experience.